Online Resource 1

Bid and award dispersion in European carbon auctions

Conditional association and composition sensitivity

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S1 Purpose and fixed numerical record

This resource documents the historical estimates and later supplementary analyses without combining them into one confirmatory study. Sections S1–S9 retain the original numerical results. Sections S10–S12 cover composition weighting, the separate 2026 sample, finite resampling diagnostics and the uncompleted price-reference analysis. Section S13 maps estimands to their evidence roles. This editorial revision changes no estimate, resample, eligibility rule or previously reported table value.

The repository is `124-creator/-123`. The input is fixed at commit `5745eb3ddca79602346053446583d794b0e09800`. The principal results are fixed at `e25c1e02e1a58e0aec1084f7419066855e9565ec`, and the robustness extension at `d4b26f1bbd8c1d7c036259e48e119c9c12f6c9a9`. The respective paths are:

- `research/pro_review_20261010/full_data/`
- `research/eex_abc_results_20261010/`
- `research/eex_robustness_20261010/`

The first path contains the raw-workbook identifiers, complete extraction, candidate sample, exclusions and cell traces. The latter two contain the analysis code, protocols, result summaries and receipts. Reproduction should start from the fixed input and rebuild the eligibility rules, not filter until the reported sample size is reached. Data rights remain with the original provider; final journal deposit or access arrangements require author confirmation.

S2 Fields, observations and eligibility

Table S1 Reported fields and research interpretation

Symbol	Original field label	Interpretation retained here
μ^b	Average volume bid per bidder	Reported bid mean
s^b	Standard deviation of bid volume per bidder	Reported bid SD; population and divisor unresolved
μ^w	Average volume won per bidder	Reported award mean
s^w	Standard deviation of volume won per bidder	Reported award SD; population and divisor unresolved
V	Auction Volume tCO2	Reported quantity, not a separately observed settlement total
B	Total Amount of Bids	Total reported bid volume
R	Cover Ratio	Reported demand coverage
N	Total Number of Bidders	Participants, not number of submitted orders
S	Number of Successful Bidders	Successful participants, not number of successful orders

Note: Quantity fields use tCO2 in the source. Literal labels are reproduced from row 6 of Primary Market Auction in each annual workbook. In particular, column M is “Total Amount of Bids” within the volume section; auction revenue is a different field. Original labels, cell positions and formats are retained in the provenance record. A reporting participant need not equal a final client or compliance installation. Field labels and arithmetic consistency do not establish identical populations for each paired mean and standard deviation.

The six annual files yield 1,327 dated observations. Restricting to T3PA, EU/DE/PL and explicit successful status yields 1,281 unique auctions. Exclusions comprise 38 other-product records, five other-series records and three cancelled auctions. Cancelled award fields remain missing rather than becoming zero dispersion. No main-sample row is removed on the basis of the size of its reported ratio.

The source audit compared 13 numeric fields for all dated observations with the extraction and the cell trace, with 17,251 comparisons in each route. An independent XML route agreed with the workbook import. The 53 candidate fields match their corresponding extraction rows; input files were unchanged after computation. These checks certify the relationship between the supplied files and the calculations, not unobserved bidder identities, every market event or historical first-release values.

S3 Projection and resampling implementation

The main nuisance space contains log cover ratio, log auction volume, log total bidders, log successful bidders, reporting-series indicators and original year–month indicators. Each auction has equal weight. Numerically stable least-squares projection gives a rank-79 pooled design with 1,202 residual degrees of freedom. The within-month implementation was cross-checked using an explicit dummy design and explicit row replication on fixed draws. Such agreement verifies a numerical route, not a second empirical sample.

For pooled block length L, draw starting months uniformly from $0, \ldots, M - L$ with $M = 72$. Append each selected contiguous block of L calendar months until at least 72 positions are obtained, then truncate. Retain all auction-series observations in each selected month. Multiplicities act as observation weights; original year–month labels are not renamed. Empty factor levels in a draw are

removed rather than represented by all-zero dummy columns. Nuisance coefficients and the focal coefficient are re-estimated in each draw, and failures are recorded without replacement draws.

The principal stream uses `SeedSequence([20261010, block_length])`. The three-month analysis uses 999 draws shared across A/B/C; the one- and six-month checks use 199 each for A and C. Quantiles are computed by linear interpolation. The prior team calculation used a different initialization; this resource retains the frozen principal intervals rather than the more favourable of alternative Monte Carlo realizations. In particular, the principal ordinary-log slope interval includes one.

For the historical split, each original half has 36 months. Independent three-month resampling within each half produces 999 paired comparisons, shared across A and C. The extension protocol records the period stream `[20261010, 2, 1]`, the reduced-control stream `[20261010, 2, 2]` and the series stream `[20261010, 2, 3]`. All nuisance slopes are separately estimated in each half. The two 97.5% slope-difference intervals constitute a Bonferroni-style sensitivity calculation only to the extent that each interval approximation is adequate. They do not correct selection of the historical question or fully represent cross-boundary dependence.

Reduced-control specifications use 499 draws, and within-series analyses use 399. Each subseries retains the full 72-month calendar axis, including months without selected auctions. EU and PL each have one such empty month. No observation is fabricated there. Singleton months contribute no net within-month information. The completed runs report no failed bootstrap draw. Bootstrap coverage remains approximate, particularly under changes in slopes or dependence.

S4 Additional numerical results

Table S2 Pooled block-length sensitivity

Spec.	Block months	Draws	β: 95% interval	θ: 95% interval
A	1	199	[0.7807, 0.9833]	[0.5903, 0.7158]
A	3	999	[0.7506, 1.0384]	[0.5714, 0.7422]
A	6	199	[0.7350, 1.0292]	[0.5564, 0.7438]
C	1	199	[0.7907, 0.9714]	[0.5855, 0.7134]
C	3	999	[0.7627, 1.0143]	[0.5660, 0.7402]
C	6	199	[0.7503, 1.0090]	[0.5526, 0.7493]

Note: These are exploratory percentile intervals. Three-month rows reproduce the principal result; they are not additional replications. Sensitivity endpoints based on 199 draws have non-negligible Monte Carlo variability.

Table S3 Influence sensitivity of residual correlation

Deletion diagnostic	A: θ range / value	C: θ range / value
One calendar month	[0.6419, 0.6656]	[0.6405, 0.6674]
One calendar year	[0.6058, 0.6981]	[0.6023, 0.6996]
One reporting series	[0.6019, 0.7120]	[0.6110, 0.7081]
One observation	[0.6467, 0.6569]	[0.6457, 0.6577]
Top 13 Cook-distance rows	0.6911	0.6930

Note: Bracketed values are deterministic deletion ranges, not confidence intervals. The largest-Cook-distance set is selected within each specification only for diagnosis. It does not define a revised data-cleaning rule. All main estimates retain 1,281 auctions.

Table S4 Within-series residual association

Spec.	Series	n	Residual df	θ [95% interval]
A	EU	840	764	0.6898 [0.6283, 0.7585]
A	DE	275	198	0.5619 [0.4567, 0.6808]
A	PL	166	90	0.8076 [0.6943, 0.8828]
C	EU	840	764	0.6839 [0.6159, 0.7587]
C	DE	275	198	0.5689 [0.4679, 0.6870]
C	PL	166	90	0.8138 [0.7005, 0.8850]

Note: Each series has its own month effects and numeric controls. Intervals use 399 three-month block draws. These are not independent-market replications or formal tests that series coefficients differ. The small PL residual sample limits interpretation.

The interquartile-scale contrasts reported in the main article are already present in the frozen extension output. Under A, $IQR\left(M_Z x\right)=0.0927037454$ and the corresponding fitted transformed-response contrast is 0.0816007553. Under C, these values are 0.1600483679 and 0.1407987656. They are partial-regression scale summaries. Exponentiating a transformed fitted value would not, without further assumptions, recover the arithmetic conditional mean of the original ratio. No such raw-scale mean or procurement outcome is estimated here.

S5 Denominator identities and rounding

For this diagnostic only, a superscript c labels constructed fields. Define $\mu^{b,c}=B/N$, $\mu^{w,c}=V/S$ and $R^c=B/V$, retaining V as the reported auction-volume field. For positive components,

$$\log r^{b,c}=\log s^b+\log N-\log R^c-\log V, \tag{S1}$$

$$\log r^{w,c}=\log s^w+\log S-\log V. \tag{S2}$$

If the common nuisance space Z^c contains these constructed log totals and counts, linear projection gives

$$M_{Z^c}\log r^{b,c}=M_{Z^c}\log s^b, M_{Z^c}\log r^{w,c}=M_{Z^c}\log s^w. \tag{S3}$$

The completed calculation yields maximum residual discrepancies of approximately 7.47×10^{-15} and 7.09×10^{-15}. These are algebraic and implementation checks, not another economic finding. They do not supply an independently observed total award W or authenticate $V=W$.

With the original reported rather than constructed values, mean discrepancies relative to B/N and V/S reach 0.5 quantity units, while the cover-ratio discrepancy relative to B/V reaches approximately 0.004994. Agreement with display precision is not certification of the historical rounding rule. The main regressions use the reported fields; neither implied means nor implied cover ratios silently replace them. The corresponding log-one-plus terms do not satisfy the additive log identities.

A related interpretation boundary concerns concentration. If m quantities genuinely form one population with common mean μ and $s^2=\sum_i\left(q_i-\mu\right)^2/\left(m-d\right)$, then its single-auction quantity-share concentration is

$$H=\frac{1}{m}+\frac{m-d}{m^2}\left(\frac{s}{\mu}\right)^2. \tag{S4}$$

This conditional moment identity is standard algebra. It does not justify substituting N or S when the standard-deviation population is unknown. Cross-auction concentration additionally requires identity links across auctions. The current study neither computes a certified bidder-population concentration index nor infers matched-population compression from its cross-auction regression.

S6 Exploration record and author responsibilities

The snapshot, target and some results were inspected before the supplementary protocols were fixed. The first-freeze record is partial. The manuscripts and intervals cannot therefore be described as preregistered, selection-adjusted or based on an untouched test sample. The principal and robustness outputs are frozen for the current revision; no result was recomputed to make the text more favourable.

An earlier supplementary protocol proposed 0.10 as an exploratory reference for the absolute standardized association. It was introduced after earlier results were known and is not an economic-cost or journal threshold. It is not used as a decision boundary in the article's figures or as proof of publication value. This presentation change leaves the numerical record unchanged.

The numerical reimplementations use the same archive and are not independent-market replications. Fitting counts are execution bookkeeping, not sample sizes. The scope of the evidence is reported-field association conditional on successful auctions. Author order and the corresponding contact have been supplied for this revision. The corresponding author has confirmed that the research received no financial support and that Xiang Wang has reviewed the manuscript. Competing interests, actual contributions, AI-use disclosure and data permissions remain separate author declarations.

S7 Interpretation of the fitted quantities

All quoted coefficients, intervals and contrasts in the article come from the frozen principal and extension outputs. Residual correlation is a symmetric in-sample statistic; the chosen direction of the regression is not evidence of temporal transmission. With one added regressor and identical residualization, partial R^2 equals the square of the residual correlation. These two numbers are not independent confirmations.

The reported fitted contrasts across the interquartile range of residualized bid dispersion remain on their respective transformed-response scales. No back-transformed conditional arithmetic mean, representative bidder outcome or monetary benefit is inferred from them. Subperiod slopes and correlations use separately estimated nuisance coefficients and residual variances. Their difference is a descriptive historical comparison, not a structural invariance test robust to arbitrary changes in reporting or covariate distributions.

S8 Descriptive context and direct correlation contrasts added to the presentation

Table S5 Descriptive statistics by historical period

Panel A: 2020–2022, n = 643

Variable	Mean	Q1	Median	Q3	Min.	Max.
Bid ratio r^b	1.532	1.345	1.520	1.712	0.874	2.622
Award ratio r^w	1.459	1.247	1.443	1.629	0.443	2.707
Cover ratio R	1.864	1.530	1.790	2.100	1.010	3.770
Auction volume V	2.692	2.363	2.515	3.082	0.972	6.399
Submitted volume B	4.844	3.887	4.761	5.588	1.937	12.899
Total bidders N	22.379	20	22	25	12	31
Successful bidders S	16.687	14	17	20	3	27

Panel B: 2023–2025, n = 638

Variable	Mean	Q1	Median	Q3	Min.	Max.
Bid ratio r^b	1.275	1.108	1.255	1.429	0.847	1.896
Award ratio r^w	1.384	1.184	1.356	1.557	0.665	2.295
Cover ratio R	1.785	1.550	1.710	1.980	1.080	3.780
Auction volume V	2.659	2.147	3.035	3.245	0.970	3.350
Submitted volume B	4.588	4.076	4.712	5.156	1.946	6.731
Total bidders N	22.948	21	23	25	15	30
Successful bidders S	17.071	15	17	20	5	27

Note: These previously tabulated summaries use the unchanged eligible auctions and equal row weights. Volume fields are in million tCO2; bidder counts and ratios are on their original scales. Quartiles use linear interpolation at (n−1)q (type 7). No p-values, distributional-equality tests, trimming or new regression are introduced. These are marginal auction-level summaries; the early and late auctions do not represent matched bidder populations. Lower marginal ratios cannot be equated with a decline in identified bidder-level inequality, and the table does not explain the change in conditional association.

The direct differences in correlation in main Table 4, Panel C, already exist in the frozen extension: `period_intervals.A.delta_theta_CI` and `period_intervals.C.delta_theta_CI` in `support/frozen_extension_results.json`, and equivalently `periods.A.delta_theta_CI95` and `periods.C.delta_theta_CI95` in the repository's `KEY_RESULTS.json`. The point differences are 0.2296942521 and 0.2305336591. These are secondary exploratory 95% intervals, not the 97.5% primary slope intervals. Displaying them now does not retrospectively change the extension protocol.

The original numeric ledgers are retained in the preceding v7 companion package. The current package supplies a newly generated display inventory in tables/DISPLAY_NUMERICS_V8.csv and a locator in REPRODUCIBILITY.md that points to fixed repository outputs. A display inventory checks editorial preservation; it does not replace raw-input verification or statistical replication. Historical descriptions and figure inputs have not been recalculated.

S9 Conditioning-set values accompanying Figure 2

Table S6 Residual association across conditioning sets

Conditioning set	A: $\hat{\theta}$ [95% interval]	C: $\hat{\theta}$ [95% interval]
Series and year–month effects	0.6243 [0.5596, 0.7067]	0.6197 [0.5494, 0.7017]
Effects + volume + total bidders	0.6289 [0.5626, 0.7094]	0.6239 [0.5521, 0.7044]
Full controls	0.6496 [0.5714, 0.7422]	0.6486 [0.5660, 0.7402]

Note: Reduced-control intervals use 499 three-month block draws. The full-control row reuses the 999-draw principal intervals. Each conditioning set defines a different descriptive projection; these are not instruments or randomized comparisons.

The table and Figure 2 use the same frozen values. The full-control row duplicates the principal estimate for orientation; it is not an additional empirical result. Figure 1 uses quartile ranges rather than inferential intervals; Figures 2 and 3 use the explicitly stated exploratory confidence intervals. These three uncertainty or dispersion objects must not be interchanged.

S10 Composition-standardization protocol and results

The supplementary protocol was recorded at commit 2151017871416fe0ad2505cfdbda23162aaa873e before the weighted fits, after the historical data and preceding results had been examined. It is not retrospective preregistration. The E1 implementation and outputs are frozen at 89d3ac3beba670838d4b62b1209f6f3412753f19. The original equal-weight estimates and intervals are retained; newly generated unweighted intervals serve paired comparisons within E1 only.

The period model has 22 columns: an intercept, four standardized log controls, four squares, two nonreference-series indicators and eleven month-of-year indicators. It excludes year and year–month identifiers. Exact selected-term mean balance is a property of the fitted unpenalized logistic overlap weights, not a new empirical finding (Li et al. 2018). It does not authenticate full distributional balance, assignment ignorability or minimum variance for a residual-correlation contrast. The outcome model is re-estimated separately in each half with original year–month effects. Early weights equal e and late weights equal 1−e; resampling multiplies these by row multiplicity.

Table S7 Weighted contrasts and paired changes

Spec.	Contrast	Estimate	95% interval	97.5% interval
A	Weighted difference	0.1971	[0.1084, 0.2910]	[0.0947, 0.3029]
A	Weighting-induced change	-0.0326	[-0.1349, 0.0802]	[-0.1520, 0.0915]
C	Weighted difference	0.1937	[0.0982, 0.2843]	[0.0847, 0.3051]
C	Weighting-induced change	-0.0368	[-0.1449, 0.0797]	[-0.1668, 0.0942]

Note: Change means the overlap-weighted minus equal-weight late-minus-early correlation difference from the same bootstrap draw, not a mediation effect. The 0.10 interpretation reference in the supplementary protocol was introduced after earlier results were known; it is neither a cost threshold nor a journal criterion. The weighted-contrast 97.5% intervals include that reference.

Table S8 Block-length sensitivity of the weighted correlation difference

Block months	Tuples sampled	A: 97.5% interval	C: 97.5% interval
1	199	[0.0991, 0.3018]	[0.0916, 0.3033]
3	999	[0.0947, 0.3029]	[0.0847, 0.3051]
6	199	[0.1049, 0.2881]	[0.0996, 0.2780]

Note: Both period and outcome models are refitted in every draw. A and C share indices. No failed draw was replaced. Short Monte Carlo runs affect endpoint precision; absence of computational failures does not establish coverage. Weight ESS values of 586.55 and 614.75 are concentration diagnostics, not independent time-sample sizes. Maximum normalized weights are approximately 0.263% and 0.314%. The failed auction-volume distribution diagnostic is shown in main Table 5, not suppressed.

S11 Additional 2026 sample and finite block sensitivity

The E3 record reports a download on 10 October 2026 at 13:33:15 UTC; E3R retrieved the same URL at 14:12:19 UTC. Both records identify 72,742 bytes and SHA-256 54e6fb649872d229333adb29b62a3ff5b7fddd7393facc047050fb8f87a1bf3d. E3R rebuilt the January–September eligibility set, excluding seven October rows from 173 dated records and retaining 166 auctions; the point estimates matched E3. This is a repeated implementation on the same snapshot, not an independent market replication, first-release audit or certification of standard-deviation populations.

Table S9 The 2026 association and short-axis diagnostic ranges

Quantity / diagnostic	A	C
Slope β	1.3137	1.2375
Residual correlation θ	0.8494	0.8350
Original 999-draw θ interval	[0.8123, 0.9149]	[0.7832, 0.9156]
Non-circular finite θ percentiles	[0.8094, 0.9149]	[0.7757, 0.9156]
Circular finite θ percentiles	[0.8064, 0.9048]	[0.7731, 0.9055]
Leave-one-month-out θ range	[0.8396, 0.8626]	[0.8230, 0.8569]

Note: Point estimates and the original 999-draw intervals belong to E3. Finite-enumeration and deletion diagnostics are post-result E3R analyses. The enumeration percentiles are 2.5% and 97.5% quantiles of equally likely ordered block triples with linear interpolation. Deletion ranges are not confidence intervals. All schemes use the same 166 auctions and fixed models; they do not provide independent market replications.

For non-circular triples, the expected original-month multiplicities are (3,6,9,9,9,9,9,6,3)/7. Endpoint underrepresentation is therefore structural, not eliminated by 999 Monte Carlo draws. The 343 ordered triples yield 84 distinct month-weight vectors. The 729 circular triples yield 163 vectors and unit expected multiplicity for every month; they also connect September to January artificially. Both full finite distributions are retained. No claim is made that the circular scheme is universally more appropriate or that exact enumeration gives exact frequentist coverage.

E3R records 2,164 primary projection requests and eight explicit-dummy numerical cross-checks, with no failures and maximum implementation difference 4.60×10^{-14}. Its six synthetic checks assess enumeration, multiplicities and projection code. These execution counts are not observations. Complete outputs are at research/eex_integration_checks_20261010/results, commit 35d8c63fc57173f67eb0bd8bb570eb289188324f; they have not been re-estimated for v8.

S12 Official-reported price comparisons and reproduction boundary

The earlier price gate concerned a reconstructed per-auction secondary-market reference. A subsequent acquisition record at commit 6cd665873c9bb11ed266bb3d5eea7e587ef2ca80 identifies a different, weaker observation layer: deviations already printed in German official auction reports. The team records 275 matched successful German EUA auctions in 24 quarters of 2020–2025, a conservative subset of 240 in 21 quarters, and 35 quarantined observations. The conservative subset contains 236 last-best-bid references and four last-trade references; 203 observations have no row-level source flags. These are acquisition-record counts, not a new independent reconstruction in this editorial revision.

The official-reported layer is potentially usable for a report-level association and does not require native ticks to answer that weaker question. However, the newly acquired row tables and source PDFs are author-held rather than present in the public handoff. No price regression has been fitted in this revision environment. The proposed response retains the signed reported percentage, converted from percentage points to basis points; a source column labelled Absolute must not automatically be treated as a modulus. Native quote age, an independently checked event clock and midpoint comparability remain unauthenticated. The one amended protocol and not-fitted integration template are separate research documents, not completed Results.

Code and aggregate outputs are fixed in the repository revisions listed in REPRODUCIBILITY.md. The dynamic 2026 source URL is https://public.eex-group.com/eex/eua-auction-report/emission-spot-primary-market-auction-report-2026-data.xlsx. The exact raw snapshot is not included in this package; temporary download files were not retained in the published diagnostic artifact. Full raw-data reproduction of that extension still requires a lawfully available matching snapshot. A source hash proves identity only when the corresponding bytes can be obtained; it is not itself an archive.

S13 Estimands and evidence status

Table S10 records the role of each analysis in the research sequence. Different transformations, uncertainty levels and later checks are not interchangeable confirmations. The labels describe recorded analysis roles, not a prospectively registered design.

Table S10 Evidence roles and limits

Analysis	Estimand	Uncertainty display	Interpretation
Historical pooled relation	Slope and residual correlation	Exploratory 95% month-block intervals	Within-sample conditional association
Historical period contrast	Primary slope difference; secondary correlation difference	97.5% slope intervals; 95% correlation intervals	Separate estimands; neither proves policy causality
Composition sensitivity	Weighted correlation difference	Exploratory 97.5% intervals; 95% paired-change intervals	Selected-moment comparison; incomplete volume-distribution balance
Additional 2026 period	Re-estimated slope and correlation	Nine-month diagnostics; finite percentile and deletion ranges	New-time description, not a frozen-model forecast
External price relation	Not estimated	Author-held official comparisons; row-level transfer and estimation pending	No price estimate; native quote reconstruction not claimed

Note: Each interval family is conditional on its approximation. No procedure adjusts the complete historical specification search, and exact enumeration does not give exact frequentist coverage. The price extension has no estimates. This table adds no statistical test.

Additional references

European Commission (2023) Auctions by the Common Auction Platform: October, November, December 2023. https://climate.ec.europa.eu/document/download/f3199005-4ca8-461b-9499-e5f3846ea4a5_en?filename=cap_report_202312_en.pdf. Accessed 10 October 2026 (parsed text, Sections 1.1.1 and Annex 2)

Li F, Morgan KL, Zaslavsky AM (2018) Balancing covariates via propensity score weighting. Journal of the American Statistical Association 113:390–400. https://doi.org/10.1080/01621459.2016.1260466